

Software Component Overview / 软件组件概述

The FRC® software consists of a wide variety of mandatory and optional components. These elements are designed to assist you in the design, development, and debugging of your robot code as well as assist with control robot operation and to provide feedback when troubleshooting. For each software component this document will provide a brief overview of its purpose, a link to the package download, if appropriate, and a link to further documentation where available.

FRC® 软件由各种必需组件和可选组件组成。这些组件旨在协助你设计、开发和调试机器人代码，也能帮助控制机器人运行，并在故障排查时提供反馈。对于每个软件组件，本文档都会简要概述其用途；在适用时提供软件包下载链接，并在有相关资料时提供进一步文档的链接。

Operating System Compatibility / 操作系统兼容性

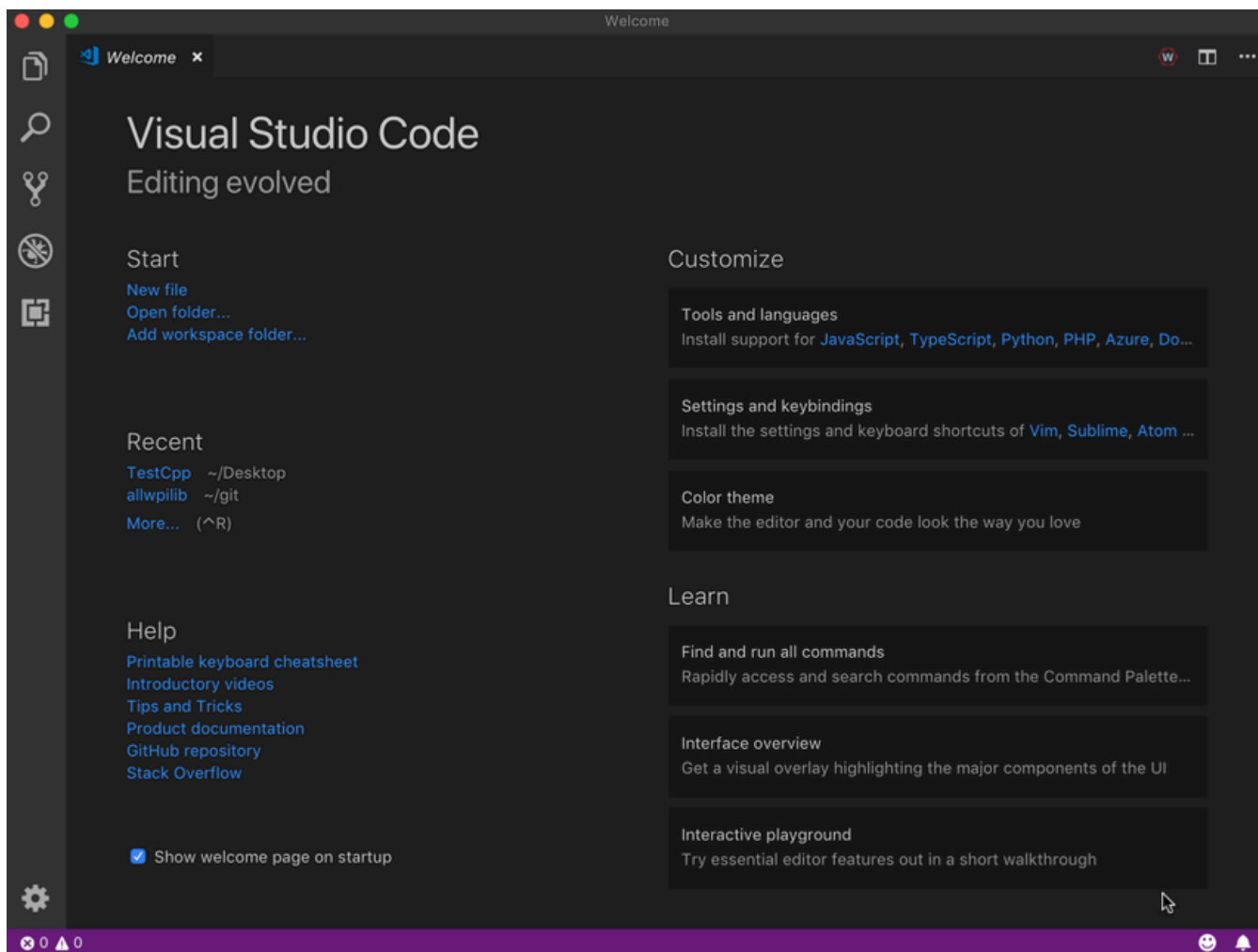
The primary supported OS for FRC components is Windows. All required FRC software components have been tested on Windows 10 & 11.

FRC 组件主要支持的操作系统是 Windows。所有必需的 FRC 软件组件均已在 Windows 10 和 11 上经过测试。

Many of the tools for Java programming are also supported and tested on macOS and Linux. Teams programming in Java should be able to develop using these systems, using a Windows system for the Windows-only operations such as the Driver Station and roboRIO Imaging Tool.

许多 Java 编程工具也支持 macOS 和 Linux，并已在这些系统上经过测试。使用 Java 编程的队伍应当能够在这些系统上进行开发，同时使用 Windows 系统执行 Driver Station 和 roboRIO Imaging Tool 等仅限 Windows 的操作。

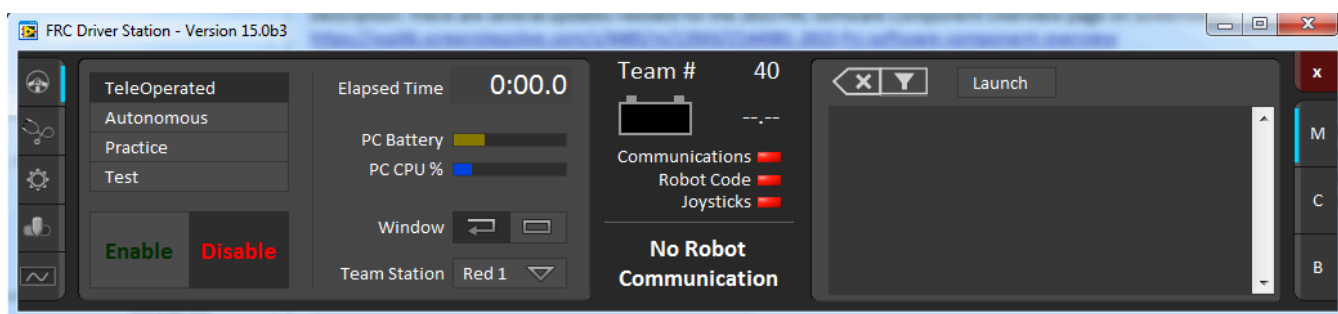
Visual Studio Code / Visual Studio Code



Visual Studio Code is the supported development environment for Java. A guide to getting started with Java for FRC, including the installation and configuration of Visual Studio Code can be found in the WPILib Installation Guide.

Visual Studio Code 是 Java 所支持的开发环境。有关开始使用 FRC Java 的指南（包括 Visual Studio Code 的安装与配置），请参阅 WPILib 文档中的“WPILib Installation Guide”。

FRC Driver Station Powered by NI LabVIEW (Windows Only) / 由 NI LabVIEW 提供支持的 FRC Driver Station（仅限 Windows）

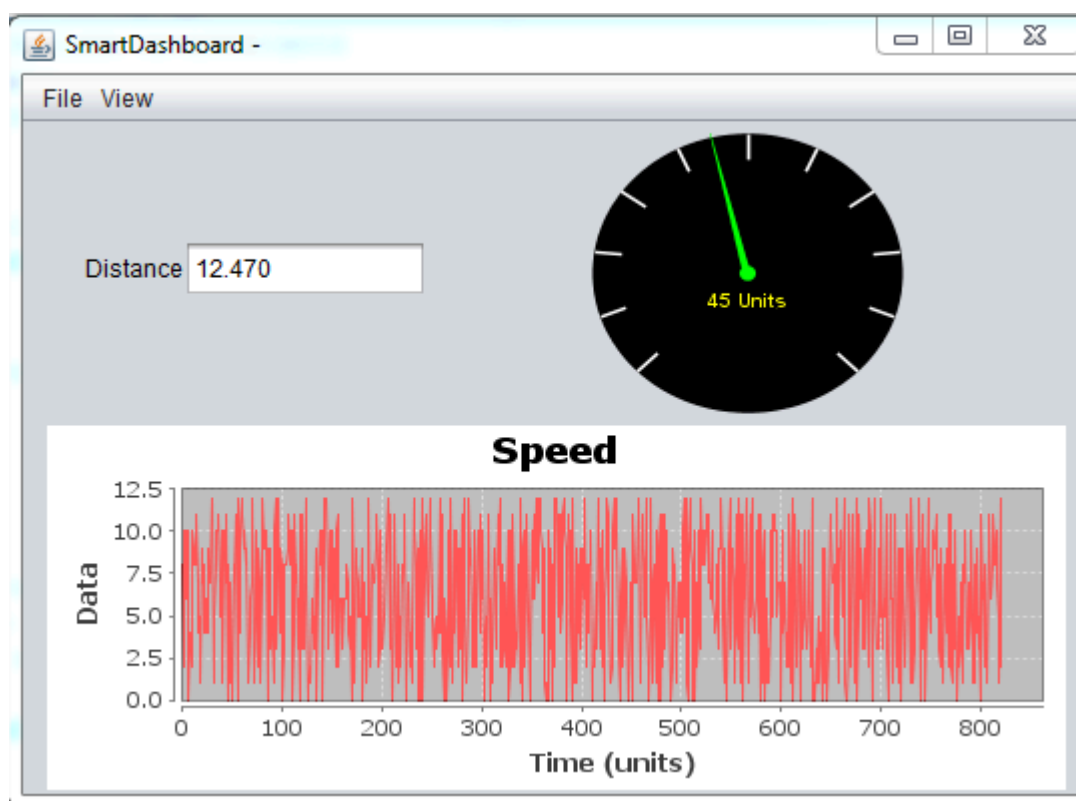


This is the only software allowed to be used for the purpose of controlling the state of the robot during competition. This software sends data to your robot from a variety of input devices. It also contains a number of tools used to help troubleshoot robot issues. More information about the FRC Driver Station Powered by NI LabVIEW can be found in the FRC Driver Station Powered by NI LabVIEW section of the WPILib documentation.

这是比赛期间唯一允许用于控制机器人状态的软件。该软件将来自各种输入设备的数据发送给机器人，其中还包含多种用于协助排查机器人问题的工具。有关由 NI LabVIEW 提供支持的 FRC Driver Station 的更多信息，请参阅 WPILib 文档中的“FRC Driver Station Powered by NI LabVIEW”章节。

Dashboard Options / Dashboard 选项

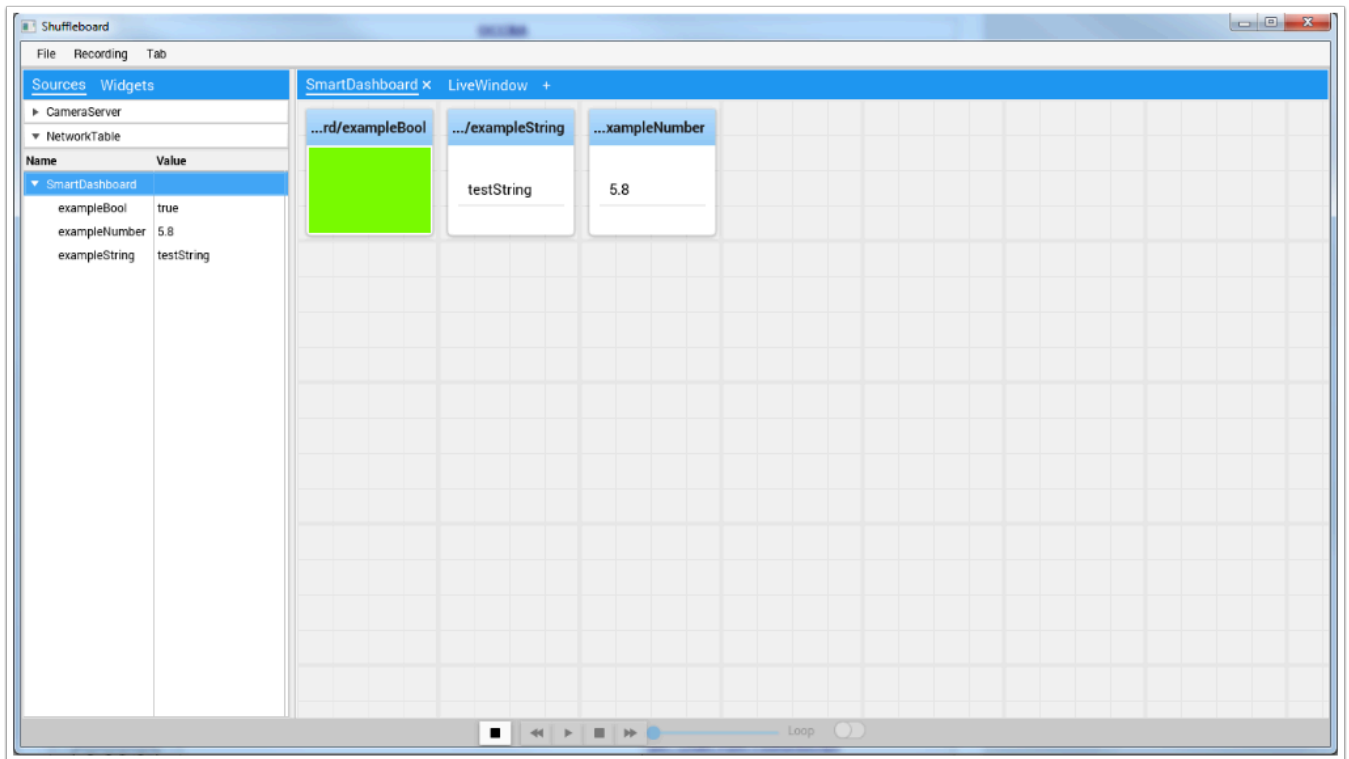
SmartDashboard / SmartDashboard



SmartDashboard allows you to view your robot data by automatically creating customizable indicators specifically for each piece of data sent from your robot. Additional documentation on SmartDashboard can be found in the SmartDashboard section of the WPILib documentation.

SmartDashboard 会为机器人发送的每项数据自动创建可自定义的指示器，使你能够查看机器人数据。有关 SmartDashboard 的更多文档，请参阅 WPILib 文档中的“SmartDashboard”章节。

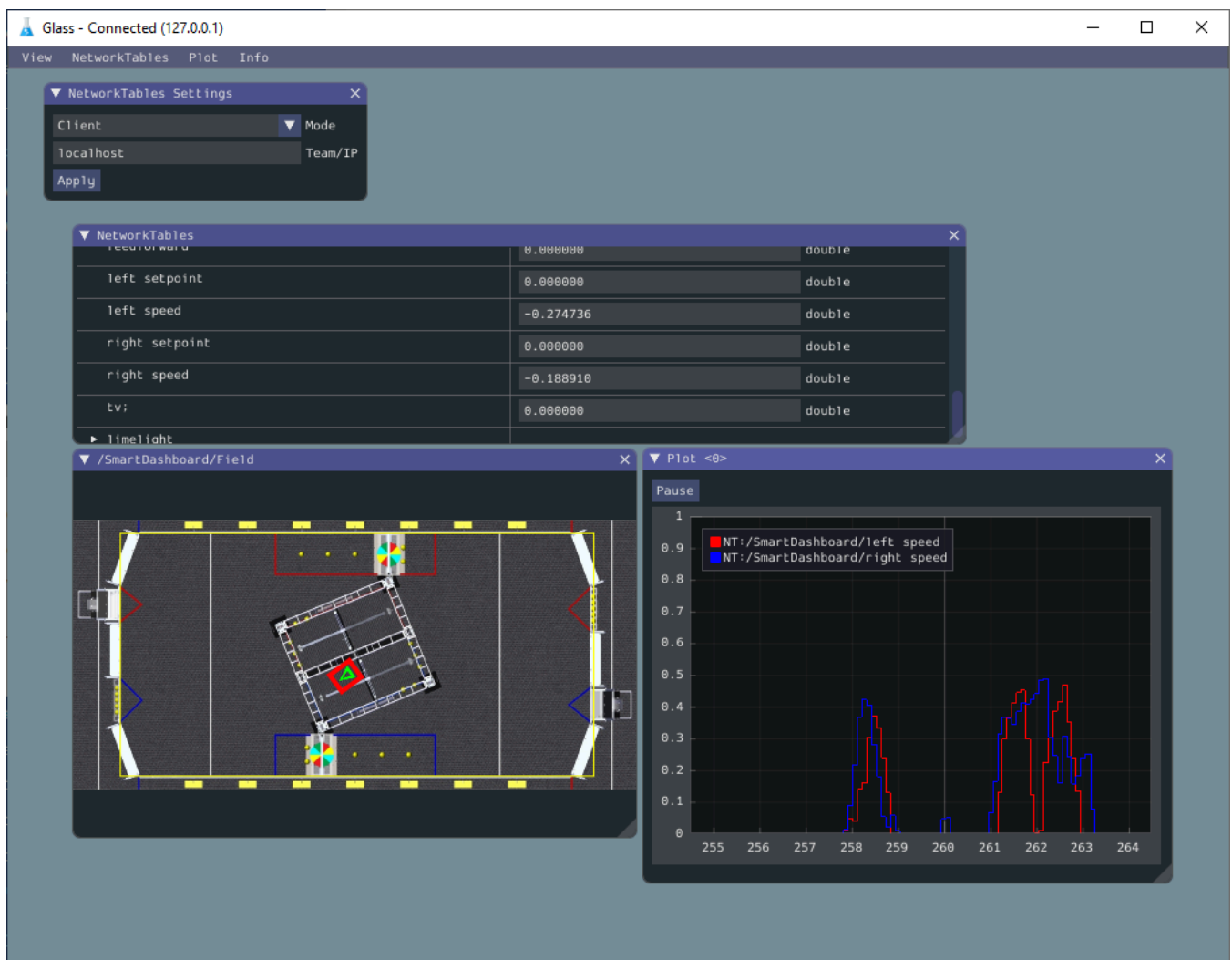
Shuffleboard / Shuffleboard



Shuffleboard has the same features as SmartDashboard. It also improves on the setup and visualization of your data with new features and a modern design at the cost of being less resource efficient. Additional documentation on Shuffleboard can be found in the Shuffleboard section of the WPILib documentation.

Shuffleboard 具有与 SmartDashboard 相同的功能。它还通过新功能和现代化设计改进了数据设置与可视化，但代价是资源使用效率较低。有关 Shuffleboard 的更多文档，请参阅 WPILib 文档中的“Shuffleboard”章节。

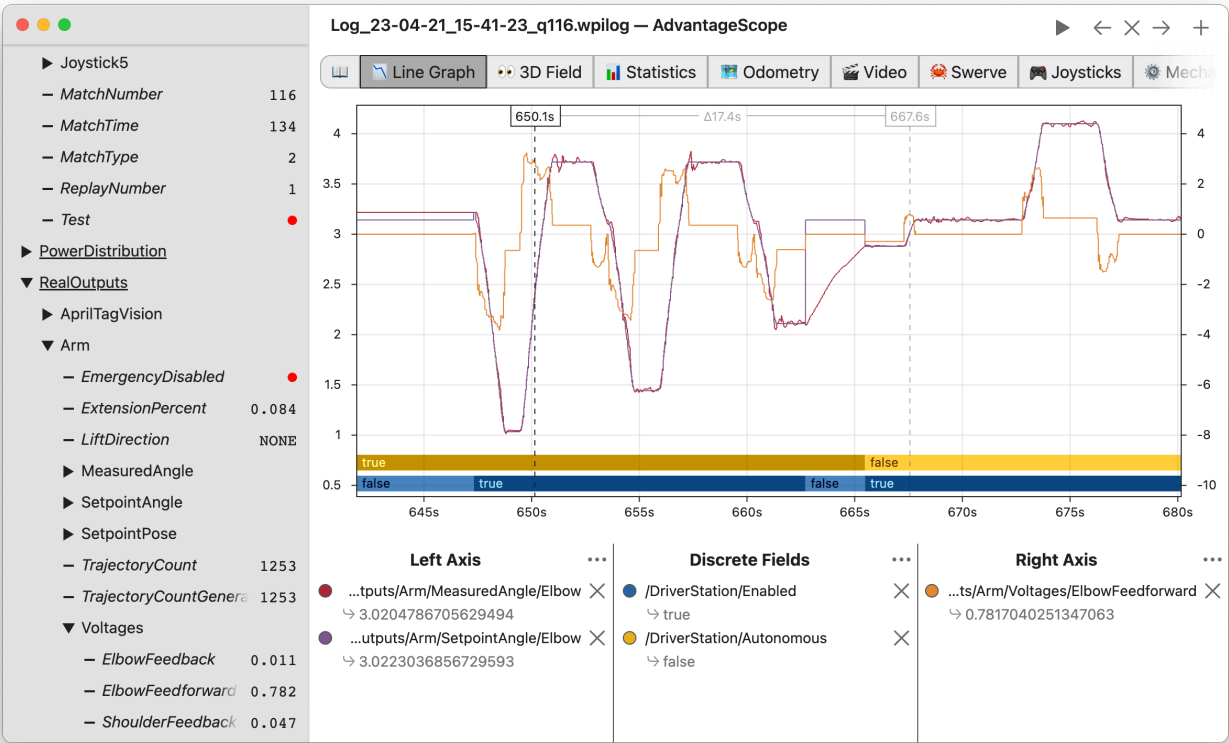
Glass / Glass



Glass is a Dashboard focused on being a programmer's tool for debugging. The primary advantages are the field view, pose visualization, and advanced signal plotting tools.

Glass 是一种专注于作为程序员调试工具的 Dashboard。它的主要优势包括场地视图、位姿 (pose) 可视化和高级信号绘图工具。

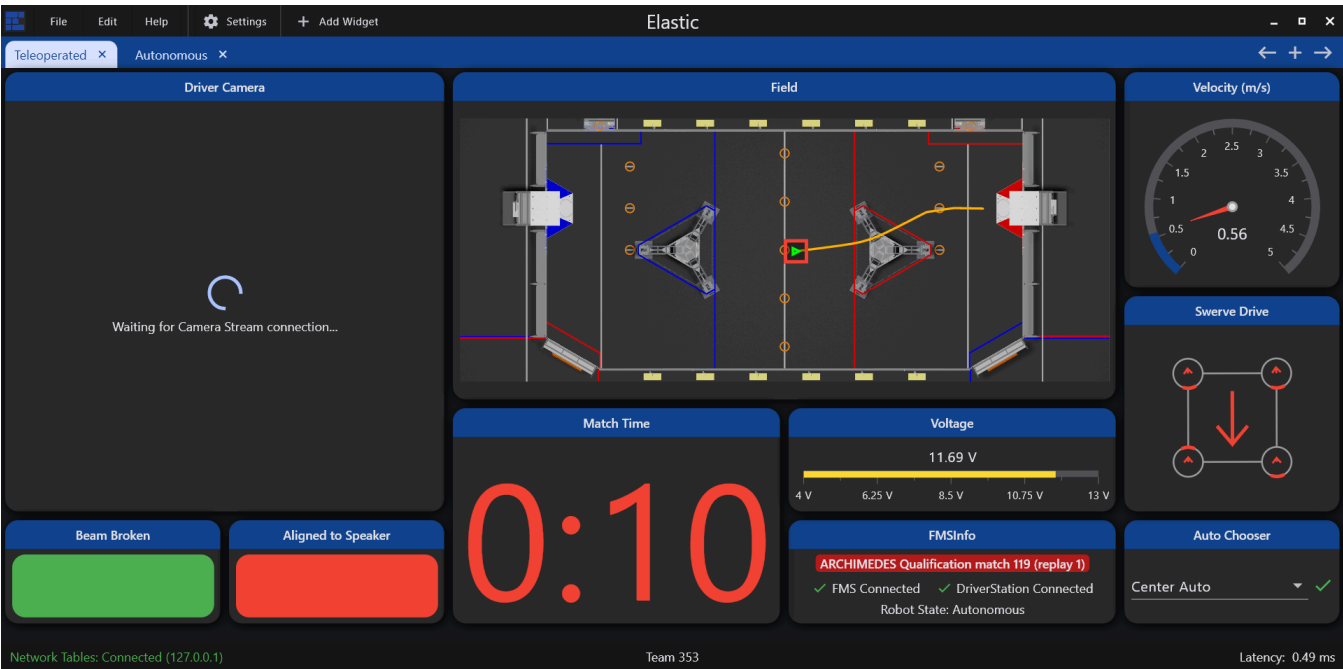
AdvantageScope / AdvantageScope



AdvantageScope is a data visualization tool for NetworkTables, WPILib data logs, and Driver Station logs. It is a programmer's tool (rather than a competition dashboard) and can be used to debug real or simulated robot code from a log file or live over the network. For more information see the AdvantageScope section in the WPILib documentation.

AdvantageScope 是用于 NetworkTables、WPILib Data Log 和 Driver Station Log 的数据可视化工具。它是一种程序员工具，而不是比赛 Dashboard；可以利用 Log File 调试真实或仿真的机器人代码，也可以通过网络实时调试。有关更多信息，请参阅 WPILib 文档中的“AdvantageScope”章节。

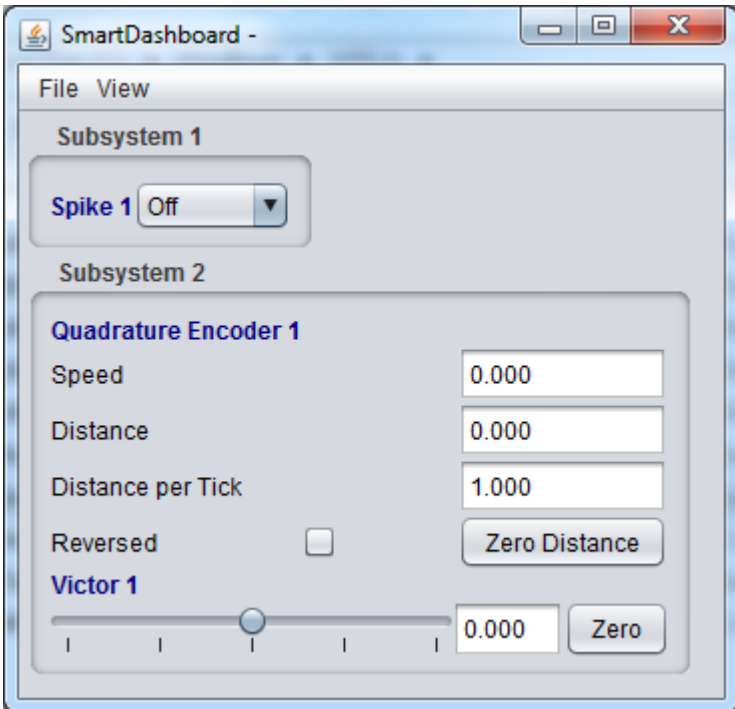
Elastic / Elastic



Elastic is a simple and modern dashboard made by Team 353. It is meant to serve as a dashboard for competition but can also be used for testing. It was built with the goal of being a drag & drop dashboard with an easy setup and elegant UI designed for a high pressure competition environment. For more information see the Elastic section in the WPILib documentation.

Elastic 是由 Team 353 制作的简洁现代 Dashboard。它旨在用作比赛 Dashboard，但也可以用于测试。其设计目标是成为一种支持拖放、设置简单并具有优雅 UI 的 Dashboard，适合高压力的比赛环境。有关更多信息，请参阅 WPILib 文档中的“Elastic”章节。

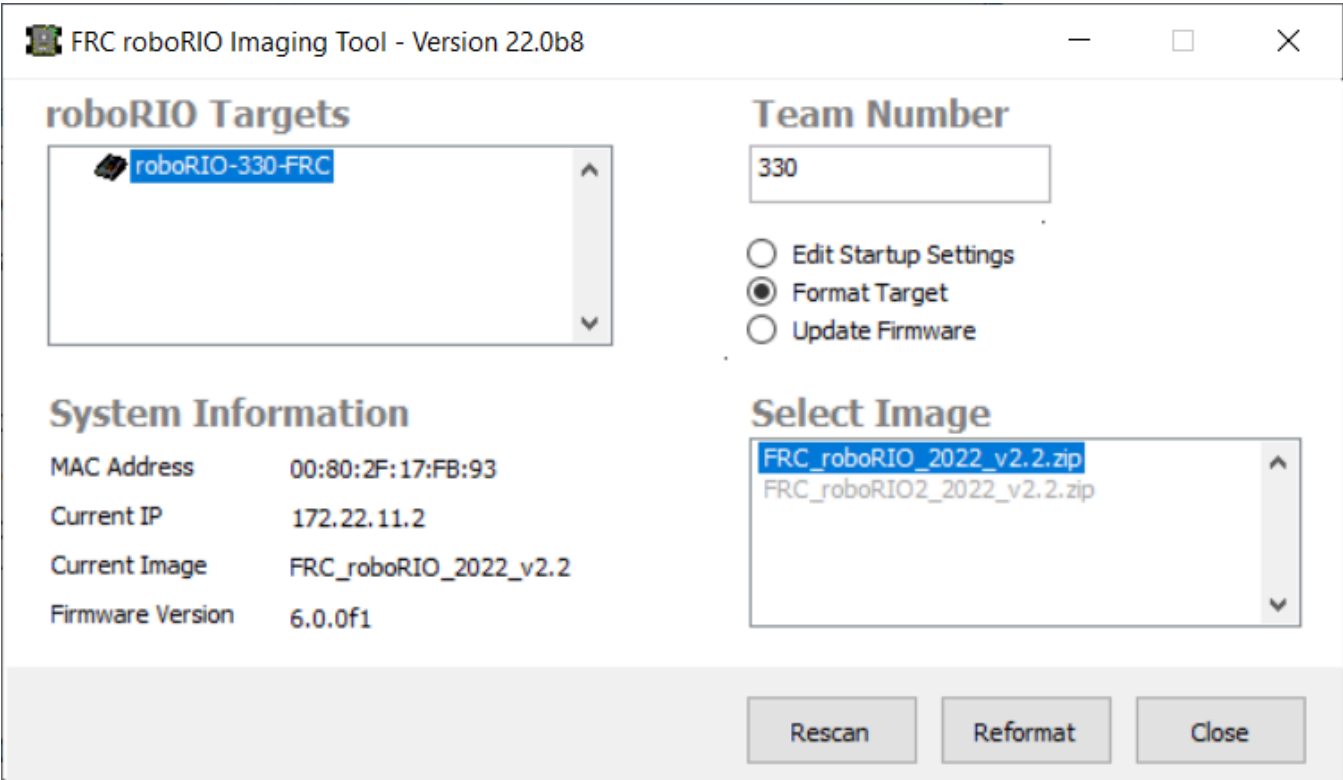
LiveWindow / LiveWindow



LiveWindow is a feature of SmartDashboard and Shuffleboard, designed for use with the Test Mode of the Driver Station. LiveWindow allows the user to see feedback from sensors on the robot and control actuators independent of the written user code. More information about LiveWindow can be found in the SmartDashboard: Test Mode and Live Window section of the WPILib documentation.

LiveWindow 是 SmartDashboard 和 Shuffleboard 的一项功能，设计用于配合 Driver Station 的 Test Mode 使用。LiveWindow 允许用户查看机器人传感器的反馈，并独立于用户编写的代码控制执行器。有关 LiveWindow 的更多信息，请参阅 WPILib 文档中的“SmartDashboard: Test Mode and Live Window”章节。

FRC roboRIO Imaging Tool (Windows Only) / FRC roboRIO Imaging Tool（仅限 Windows）



This tool is used to format and setup a roboRIO for use in FRC. Installation instructions can be found in Installing the FRC Game Tools in the WPILib documentation. Additional instructions on imaging your roboRIO using this tool can be found in Imaging your roboRIO in the WPILib documentation.

该工具用于格式化和设置 roboRIO，使其能够用于 FRC。安装说明请参阅 WPILib 文档中的“Installing the FRC Game Tools”。使用该工具为 roboRIO 写入镜像的其他说明，请参阅 WPILib 文档中的“Imaging your roboRIO”。

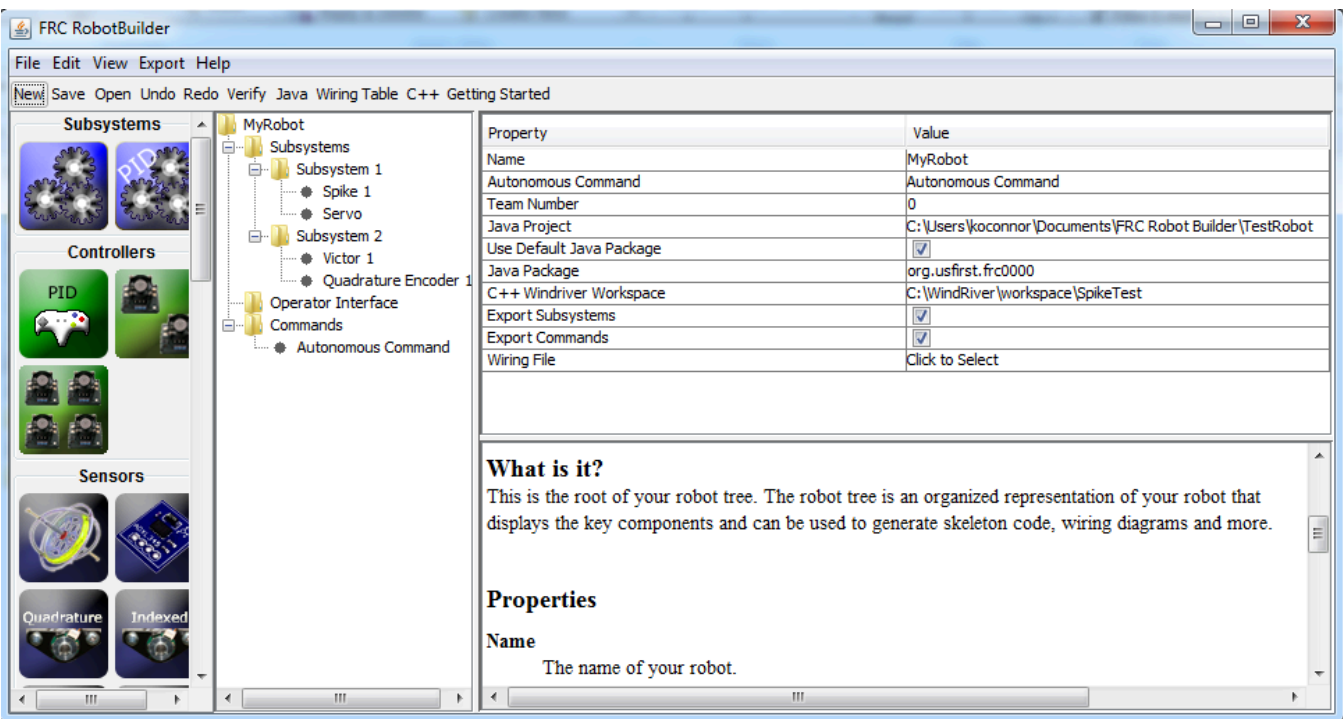
FRC Driver Station Log Viewer (Windows Only) / FRC Driver Station Log Viewer (仅限 Windows)



The FRC Driver Station Log Viewer is used to view logs created by the FRC Driver Station. These logs contain a variety of information important for understanding what happened during a practice session or FRC match. More information about the FRC Driver Station Log Viewer and understanding the logs can be found in the Driver Station Log File Viewer section of the WPILib documentation.

FRC Driver Station Log Viewer 用于查看 FRC Driver Station 创建的 Log。这些 Log 包含多种信息，对于理解练习过程或 FRC 比赛期间发生的情况非常重要。有关 FRC Driver Station Log Viewer 以及如何理解这些 Log 的更多信息，请参阅 WPILib 文档中的“Driver Station Log File Viewer”章节。

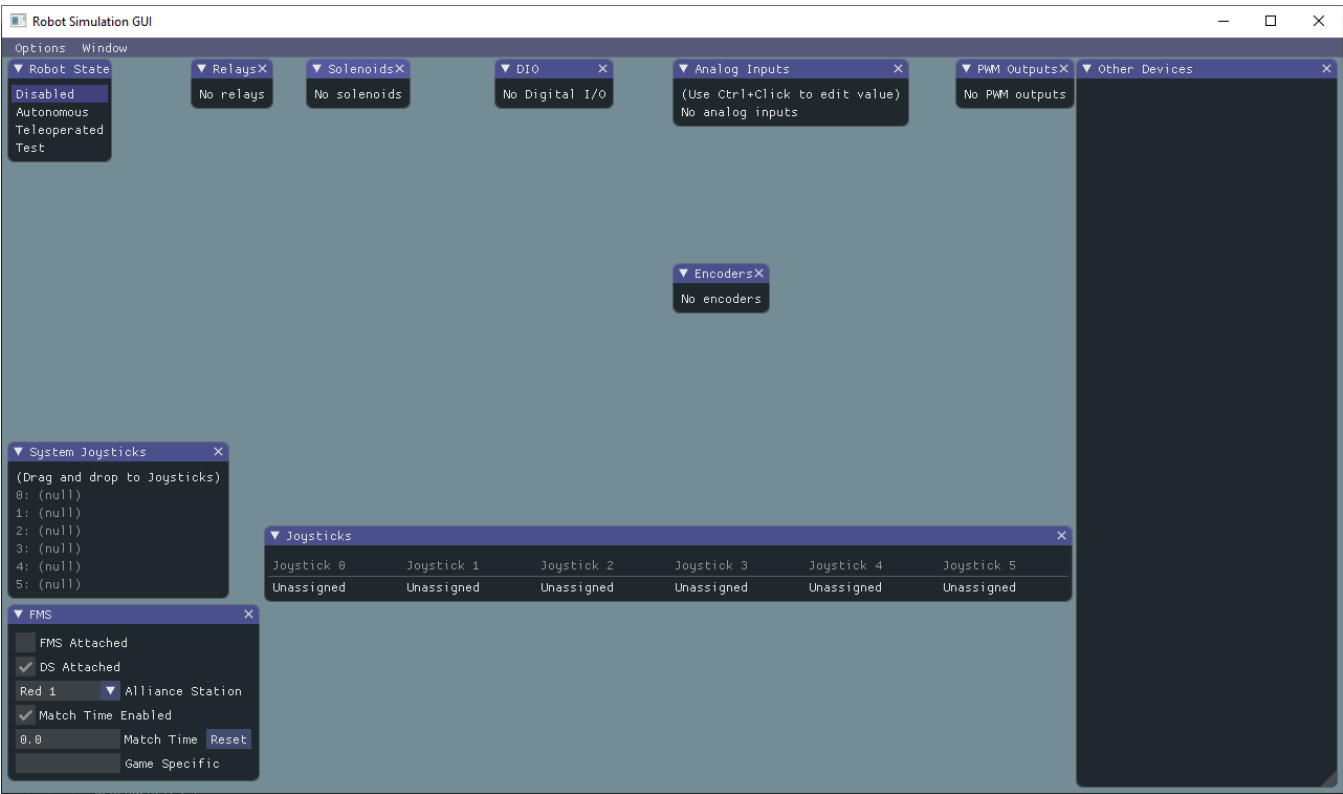
RobotBuilder / RobotBuilder



RobotBuilder is a tool designed to aid in setup and structuring of a Command Based robot project for Java. RobotBuilder allows you to enter in the various components of your robot subsystems and operator interface and define what your commands are in a graphical tree structure. RobotBuilder will then generate structural template code to get you started. More information about RobotBuilder can be found in the RobotBuilder section of the WPILib documentation. More information about the Command Based programming architecture can be found in the Command-Based Programming section of the WPILib documentation.

RobotBuilder 是一种用于协助设置和组织 Java Command-Based 机器人项目的工具。RobotBuilder 允许你输入机器人各 subsystem 和 Operator Interface 的组件，并在图形化树状结构中定义 Command。随后，RobotBuilder 会生成结构化模板代码，帮助你开始开发。有关 RobotBuilder 的更多信息，请参阅 WPILib 文档中的“RobotBuilder”章节。有关 Command-Based 编程架构的更多信息，请参阅 WPILib 文档中的“Command-Based Programming”章节。

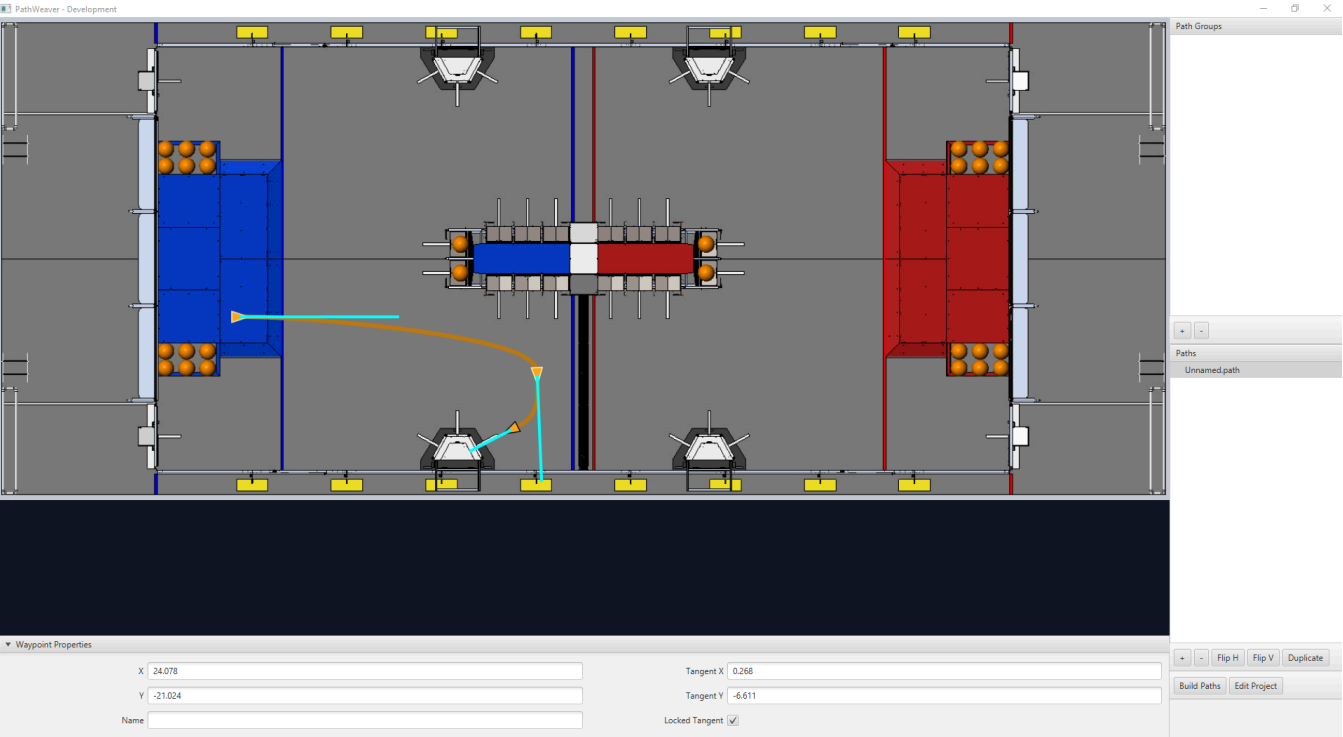
Robot Simulation / 机器人仿真



Robot Simulation offers a way for Java teams to verify their actual robot code is working in a simulated environment. This simulation can be launched directly from VS Code and includes a 2D field that users can visualize their robot's movement on. For more information see the Robot Simulation section in the WPILib documentation.

Robot Simulation 为 Java 队伍提供了一种方法，用于验证实际机器人代码能否在仿真环境中工作。可以直接从 VS Code 启动该仿真；仿真中包含一个 2D 场地，用户可以在其中可视化机器人的运动。有关更多信息，请参阅 WPILib 文档中的“Robot Simulation”章节。

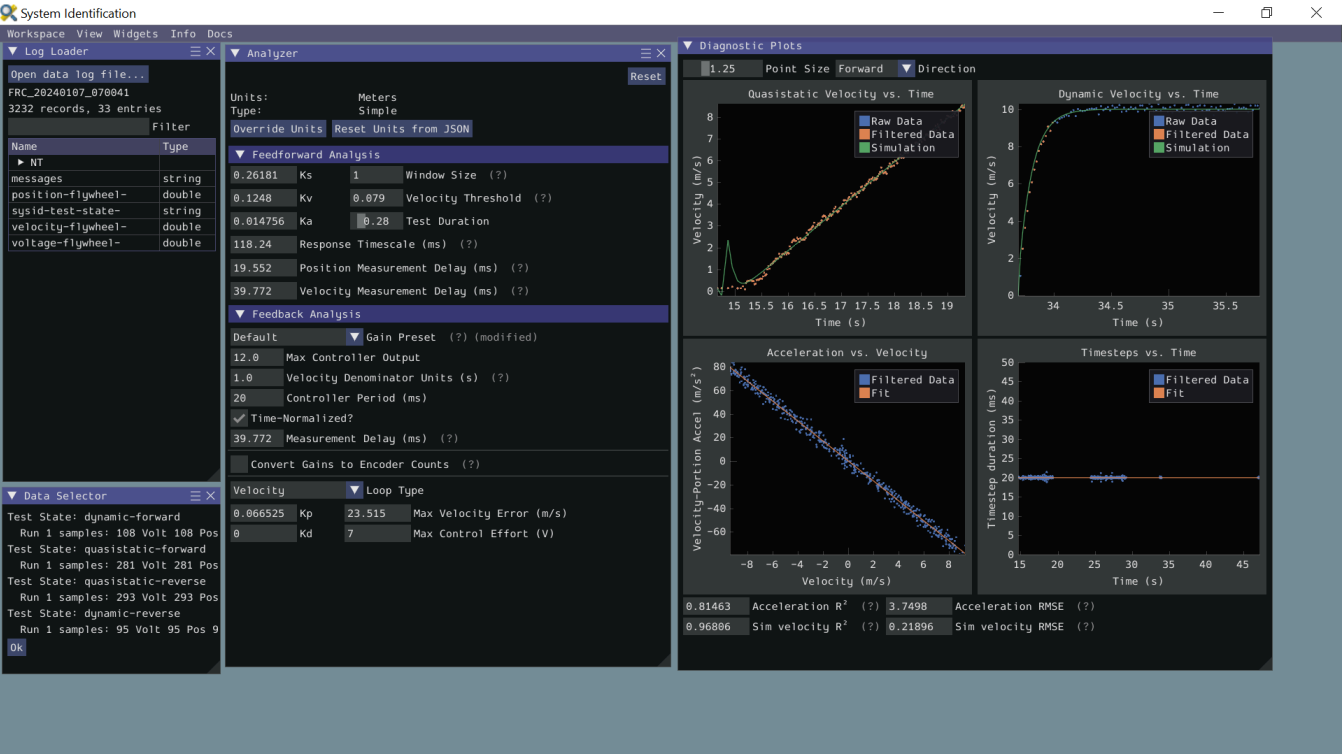
PathWeaver / PathWeaver



PathWeaver allows teams to quickly generate and configure paths for advanced autonomous routines for differential drivetrains. These paths have smooth curves allowing the team to quickly navigate their robot between points on the field. For more information see the PathWeaver section in the WPILib documentation.

PathWeaver 允许队伍为 Differential Drivetrain 的高级 Autonomous Routine 快速生成和配置路径。这些路径具有平滑曲线，使队伍能够让机器人在场地上的各点之间快速移动。有关更多信息，请参阅 WPILib 文档中的“PathWeaver”章节。

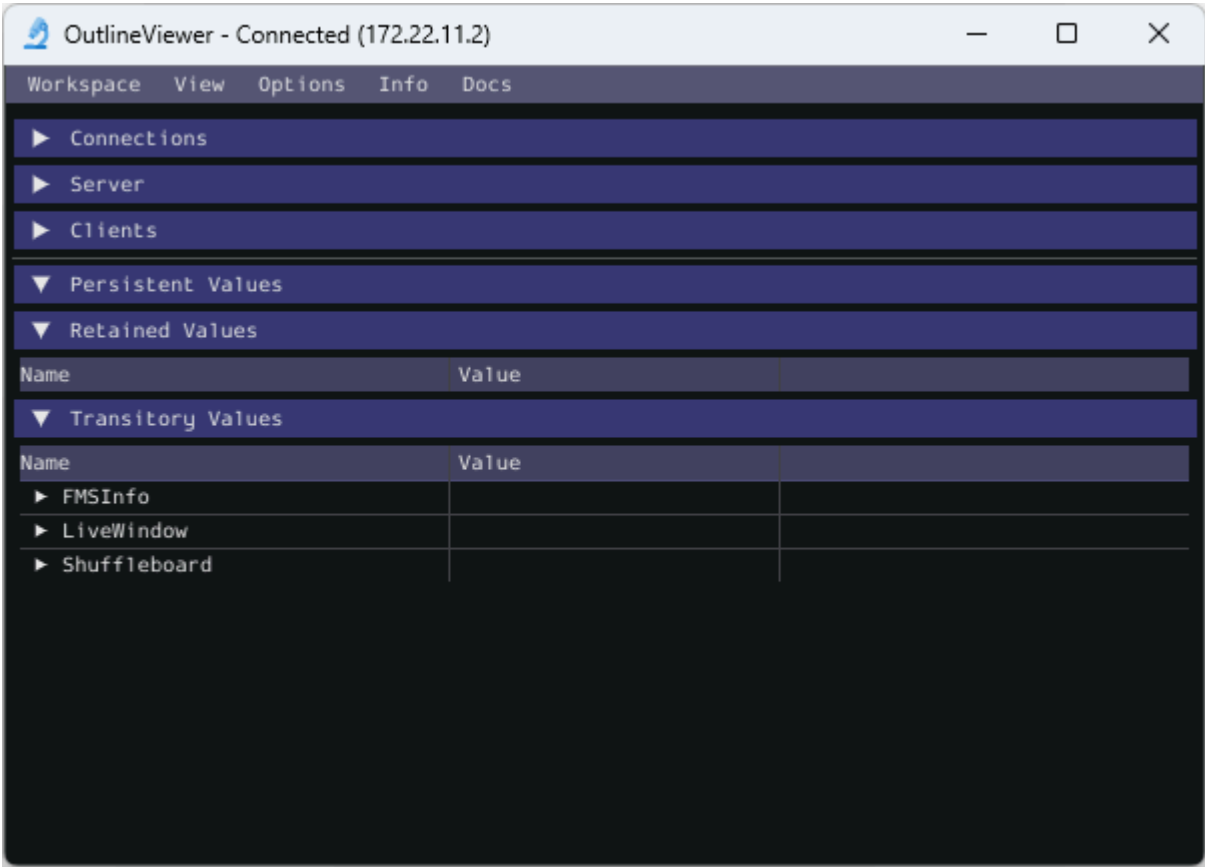
System Identification / 系统辨识



This tool helps teams automatically calculate constants that can be used to describe the physical properties of your robot for use in features like robot simulation, trajectory following, and PID control. For more information see the System Identification section in the WPILib documentation.

该工具可帮助队伍自动计算用于描述机器人物理特性的常数，以便在机器人仿真、轨迹跟踪和 PID 控制等功能中使用。有关更多信息，请参阅 WPILib 文档中的“System Identification”章节。

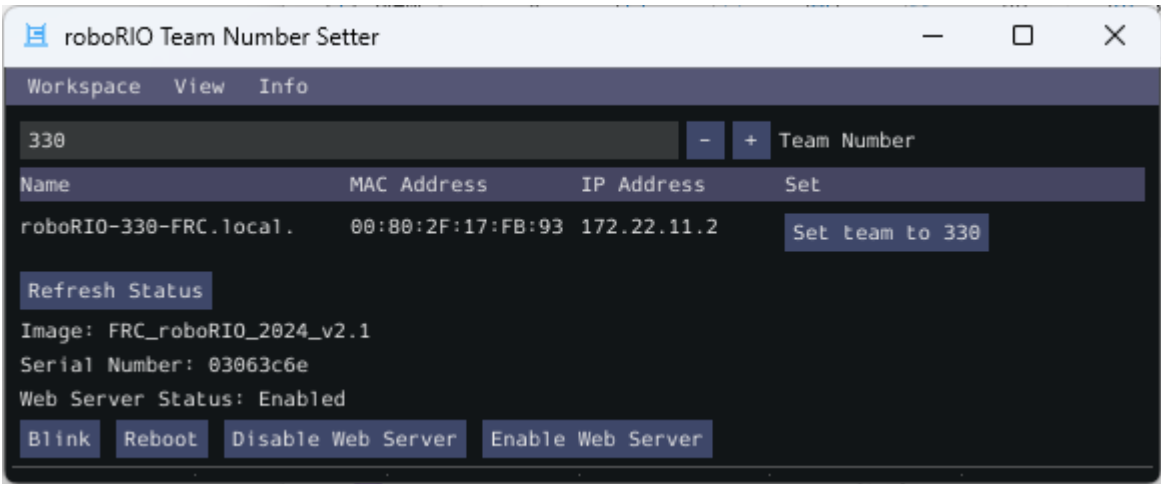
OutlineViewer / OutlineViewer



OutlineViewer is a utility used to view, modify, and add to all of the contents of the NetworkTables for debugging purposes. For more information see the Outline Viewer section in the WPILib documentation.

OutlineViewer 是一种实用工具，用于查看、修改 NetworkTables 中的全部内容并向其中添加内容，以便进行调试。有关更多信息，请参阅 WPILib 文档中的“Outline Viewer”章节。

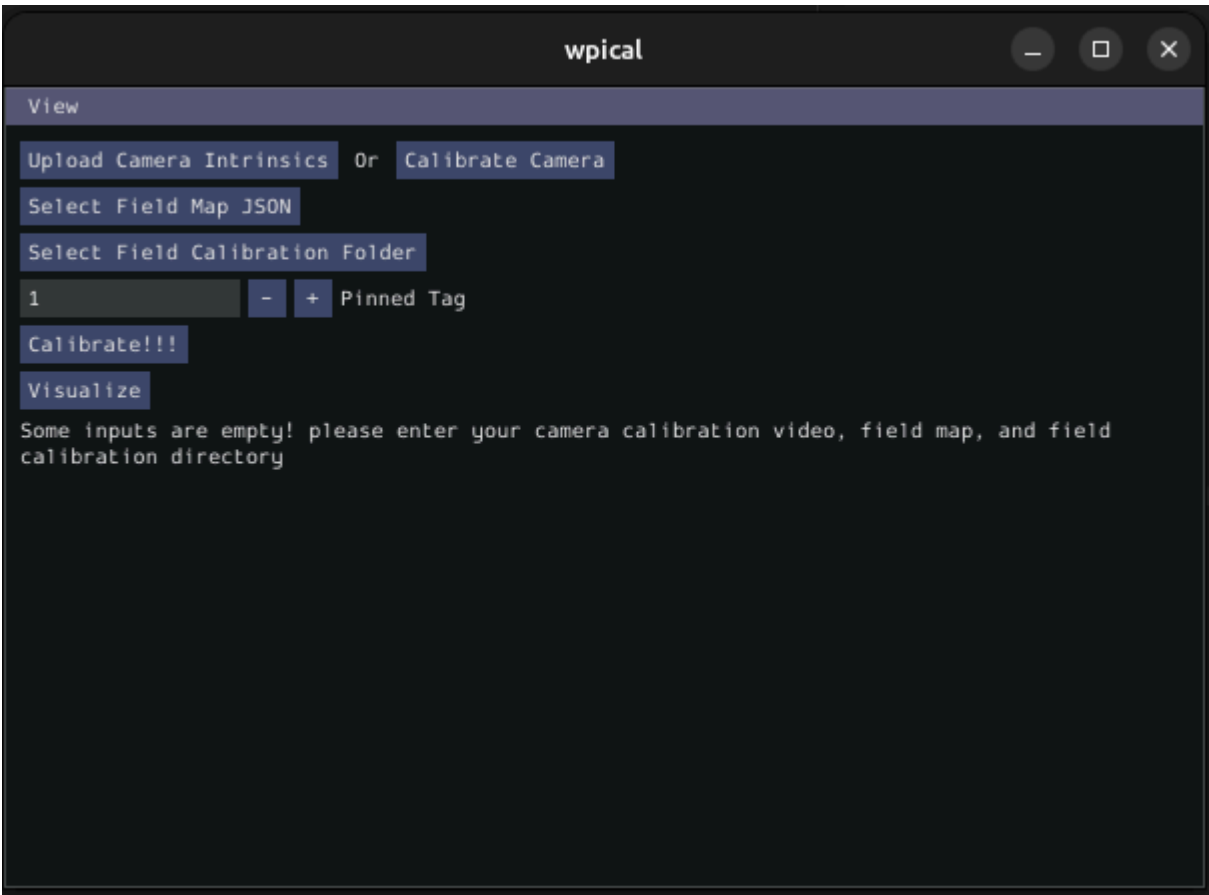
roboRIO Team Number Setter / roboRIO Team Number Setter



The roboRIO Team Number Setter is a cross-platform utility that can be used to set the team number on the roboRIO. It is an alternative to the roboRIO imaging tool for setting the team number. For more information see the roboRIO Team Number Setter section in the WPILib documentation.

roboRIO Team Number Setter 是一种跨平台实用工具，可用于设置 roboRIO 上的队伍编号。它是使用 roboRIO Imaging Tool 设置队伍编号的替代方案。有关更多信息，请参阅 WPILib 文档中的“roboRIO Team Number Setter”章节。

WPICAL / WPICAL



WPICal is a cross-platform utility that can be used to calibrate the Apriltags on FRC fields to correct for field setup error. For more information see the WPICal section in the WPILib documentation.

WPICal 是一种跨平台实用工具，可用于校准 FRC 场地上的 AprilTag，以修正场地搭建误差。有关更多信息，请参阅 WPILib 文档中的“WPICal”章节。